SE302: Advanced Software Architecture

Term: Fall 2026 | **Professor:** Dr. Robert Hayes | **Dates:** Sep 15 - Dec 20, 2026

Important Note on Assessment Dates

NOTICE FOR STUDENTS: Several examination dates and team milestone deadlines are currently **TBA** pending university room scheduling committee approval. Please verify dates once announced.

Assessment Schedule (With Pending / TBA Items)

Assessment Name	Type	Due Date	Weight	Status / Notes
Assignment 1: Microservices Design	Assignment	2026-10-02	15%	Confirmed
Midterm Exam (In-Class)	Midterm	TBA	20%	Room scheduling pending (TBA)
Group Project Milestone 1	Project	2026-11-10	15%	Confirmed
Group Project Milestone 2	Project	TBA	15%	Date to be announced
Research Presentation	Presentation	2026-12-05	10%	Confirmed
Final Comprehensive Examination	Final	TBA	25%	Pending Registrar (TBA)

Course Outline

- Module 1: Monolith to Distributed Systems Decomposition
- Module 2: Event-Driven Architecture & Message Brokers
- Module 3: Distributed Transactions, Saga Pattern, and CAP Theorem
- Module 4: API Gateways, Service Meshes, and Observability
- Module 5: Resiliency Patterns (Circuit Breakers, Bulkheads, Retries)